Introduction to Biophysics II: Quantum physics & biology

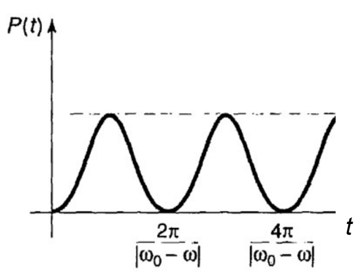
Lecture 16:
Quantum coherence in biology - I

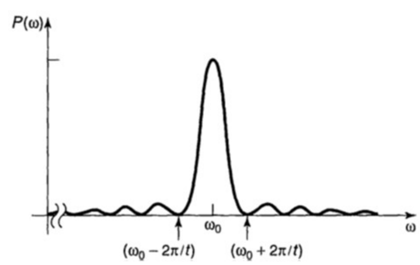
Reading:
Modern optical spectroscopy, chapter 10

Transition between states

Lecture 3

$$P_{a\to b}(t) = |c_b(t)|^2 \cong \frac{|V_{ab}|^2}{\hbar^2}\frac{\sin^2[(\omega_0-\omega)t/2]}{(\omega_0-\omega)^2}.$$



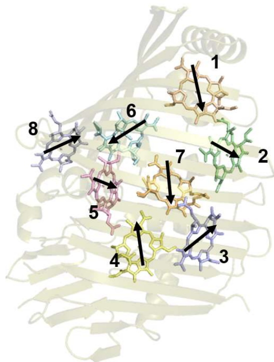


Maximizes at $\omega=\omega_0$
Note: sin(x)/x=1 when x→0

- When we keep perturbation (excitation) ON, the system oscillates between two states.
- Oscillation frequency depends on a mismatch in transition frequency and light frequency

Quantum coherence

When 2 (or more) states of one system are coupled, and there is a frequency mismatch, quantum beats can arise – oscillations in measured values.
Example: Fenna Matthews Olson complex



First paper on quantum coherence in photosynthesis:

ELSEVIER

Chemical Physics 223 (1997) 303–312

Chemical Physics

Oscillating anisotropies in a bacteriochlorophyll protein: Evidence for quantum beating between exciton levels

Sergei Savikhin, Daniel R. Buck, Walter S. Struve *

Ames Laboratory-USDOE and Department of Chemistry, Iowa State University, Ames, IA 50011, USA

Received 31 March 1997

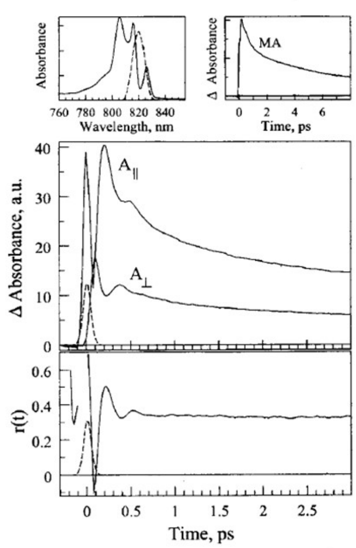
3

Quantum coherence

Example: Fenna Matthews Olson complex

Two excitonic bands at 825 and 815 nm have nearly perpendicular transition dipole moments. One may think of these as two oscillating dipoles, when the polarized pump is spectrally broad and excites, a superposition of excited states is created at time 0 with a net transition dipole directed between these two.

Since these dipoles oscillate at different frequencies, the direction of the sum will be rotating at the frequency defined by a mismatch in energies of transitions, $\Delta E/\hbar$. When probed with polarized light with two orthogonal polarizations (|| and $\perp$ to pump), signals will oscillate out of phase, their difference (anisotropy) will thus also oscillate.



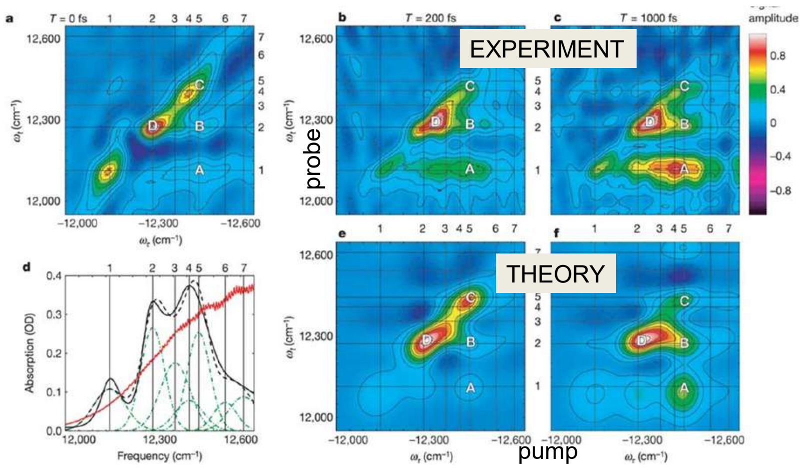
4

Quantum coherence

Two-dimensional spectroscopy of electronic couplings in photosynthesis

Tobias Brixner, Jens Stenger, Harsha M. Vaswani, Minhaeng Cho, Robert E. Blankenship & Graham R. Fleming

Nature **434**, 625–628(2005) | Cite this article

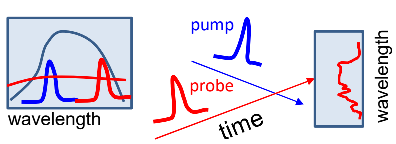


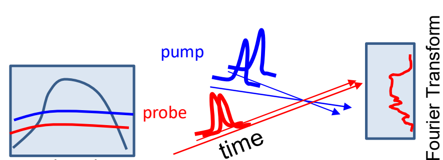
2D electronic spectroscopy

A variant of a pump-probe time-resolved spectroscopy

Difference:

	Traditional p-p	2D spectroscopy
Pump	single wavelength	wide spectrum: Fourier transform
Probe	single wavelength wide spectrum: (monochromator + CCD)	wide spectrum: monochromator + CCD or Fourier transform



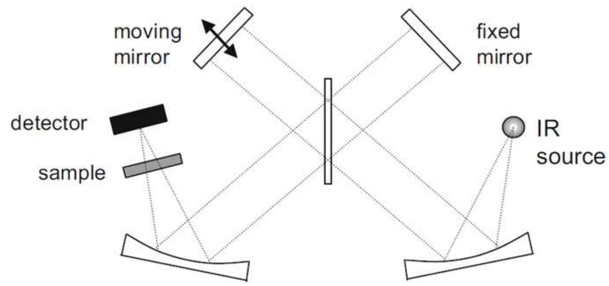


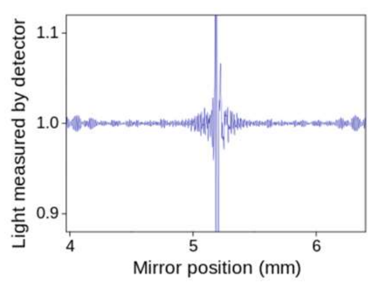
Variable time delay between two pump pulses: interference → Fourier transform → wavelength

Recall: Fourier spectrometer

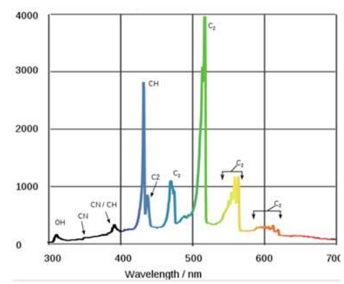
Lecture 6

Fourier spectrometer:
works well in IR





Fourier transform

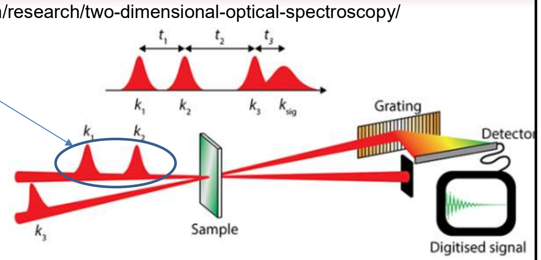


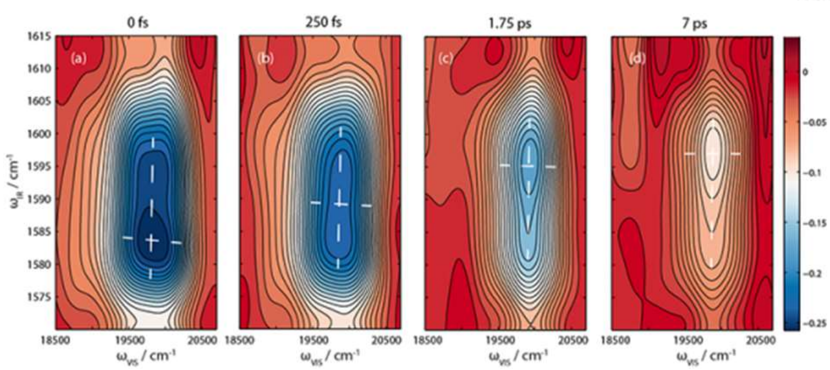
7

2D electronic spectroscopy

https://bristoldynamics.com/research/two-dimensional-optical-spectroscopy/

Time delay between two pump pulses: interference pattern
Fourier: → time to energy (wavelength) conversion





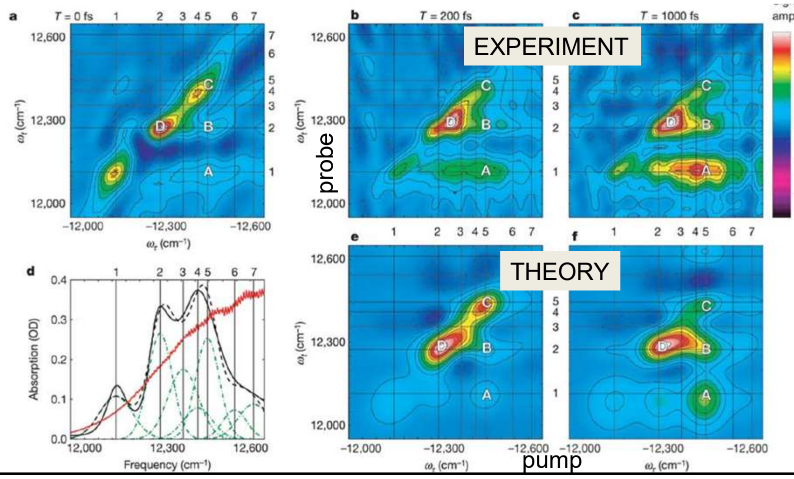
8

Quantum coherence

Two-dimensional spectroscopy of electronic couplings in photosynthesis

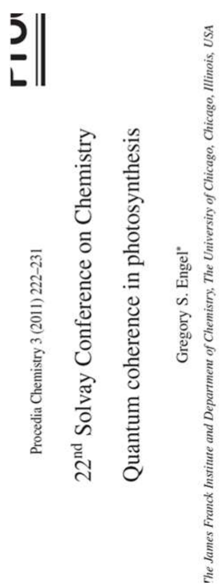
Tobias Brixner, Jens Stenger, Harsha M. Vaswani, Minhaeng Cho, Robert E. Blankenship & Graham R. Fleming

Nature **434**, 625–628(2005) | Cite this article



9

Quantum coherence

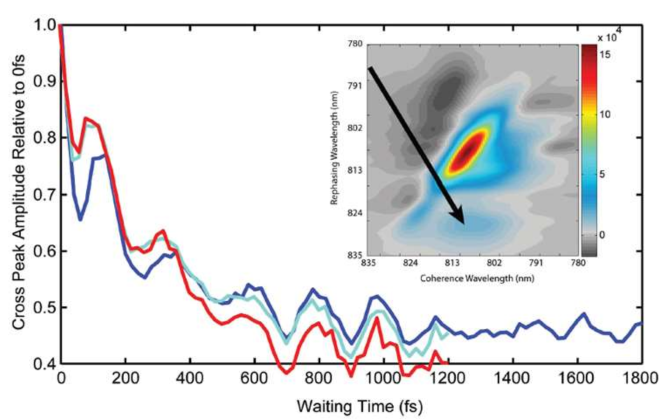

Procedia Chemistry 3 (2011) 222–231

22nd Solvay Conference on Chemistry

Quantum coherence in photosynthesis

Gregory S. Engel*

The James Franck Institute and Department of Chemistry, The University of Chicago, Chicago, Illinois, USA

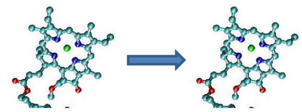


Quantum beating indicates long-lived coherence. Two-dimensional spectra of the Fenna-Matthews-Olson complex show clear beating signals in the cross-peak between excitons 1 and 2 as a function of waiting time. The three beating traces represent three replicates prepared and sampled independently.

10

Recall: Förster/Dexter approach

Donor Acceptor



Hamiltonian: $\tilde{\mathrm{H}} = \tilde{\mathrm{H}}_1 + \tilde{\mathrm{H}}_2 + \tilde{\mathrm{H}}_{21}$

interaction

Supermolecular state:

Recall Lectures 3, 6

Excitation on donor

Excitation on acceptor

$$\Psi_B = C_1\psi_1 + C_2\psi_2 = C_1\phi_{1b}\chi_{1b}\phi_{2a}\chi_{2a} + C_2\phi_{1a}\chi_{1a}\phi_{2b}\chi_{2b}$$

Solve time-dependent SE: $\hat{H}\Psi = i\hbar\frac{\partial\Psi}{\partial t}$

Perturbation theory: interaction is small

At time 0 C_1=1, C_2=0

Solution (1st order):

$$\partial C_1/\partial t = -(i/\hbar)\left\langle\psi_1\middle|\tilde{\mathrm{H}}'\middle|\psi_2\right\rangle\exp[i(E_1-E_2)t/\hbar]C_2$$

$$\partial C_2/\partial t = -(i/\hbar)\left\langle\psi_2\middle|\tilde{\mathrm{H}}'\middle|\psi_1\right\rangle\exp[i(E_2-E_1)t/\hbar]C_1$$

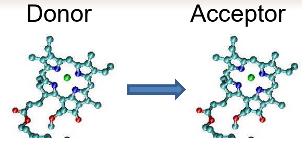
$$C_2(t) = -\frac{i}{\hbar}\tilde{H}_{21}\frac{\exp(i\omega_0 t)-1}{i\omega_0} \qquad \omega_0 \equiv (E_2-E_1)/\hbar$$

Problem: This was the first-order approximation solution, C_1=1 (time-independent)
We neglected the possibility that the system would return to its initial state
We assumed that vibrational relaxation is fast compared to energy transfer, i.e., once energy is transferred, there is no resonance between the two states.

11

Accounting for coherence

Donor Acceptor



Hamiltonian: $\tilde{\mathrm{H}} = \tilde{\mathrm{H}}_1 + \tilde{\mathrm{H}}_2 + \tilde{\mathrm{H}}_{21}$

interaction

More general solution:
Still use

ignored earlier

$$\partial C_1/\partial t = -(i/\hbar)\left\langle\psi_1\middle|\tilde{\mathrm{H}}'\middle|\psi_2\right\rangle\exp[i(E_1-E_2)t/\hbar]C_2 - (i/\hbar)\boxed{\left\langle\psi_1\middle|\tilde{\mathrm{H}}'\middle|\psi_1\right\rangle}C_1$$

$$\partial C_2/\partial t = -(i/\hbar)\left\langle\psi_2\middle|\tilde{\mathrm{H}}'\middle|\psi_1\right\rangle\exp[i(E_2-E_1)t/\hbar]C_1 - (i/\hbar)\boxed{\left\langle\psi_2\middle|\tilde{\mathrm{H}}'\middle|\psi_2\right\rangle}C_2$$

Shift of energies due to interaction

If $\langle\psi_i|H'|\psi_i\rangle$ are small (and usually they are), we can omit these terms, and the solution is simple (as we did in Lecture 3)

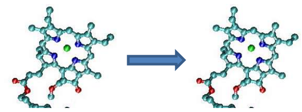
If not: redefine $H_{ii} = \left\langle\psi_i\,|\,\tilde{H}_i + \tilde{H}'\,|\,\psi_i\right\rangle = E_i + \left\langle\psi_i\,|\,\tilde{H}'\,|\,\psi_i\right\rangle$

→ Get the same simple equation

Will now include energy shift before we wrote $H_{ii}=E_i$

12

Time dependent SE: solution



We have: $\partial C_1/\partial t = -(i/\hbar)H_{12}\exp(iE_{12}t/\hbar)C_2$

$$\partial C_2/\partial t = -(i/\hbar)H_{21}\exp(iE_{21}t/\hbar)C_1$$

$$H_{21} = \left\langle \psi_2 \middle| \tilde{\mathrm{H}}' \middle| \psi_1 \right\rangle \quad (=H_{12})$$

$$E_{21} = H_{22} - H_{11} = E_2 + \left\langle \psi_2 \,|\, \tilde{H}' \,|\, \psi_2 \right\rangle - E_1 - \left\langle \psi_1 \,|\, \tilde{H}' \,|\, \psi_1 \right\rangle$$

To solve: differentiate over time again

$$\frac{\partial^2 C_1}{\partial t^2} = -\frac{i}{\hbar}H_{21}\frac{iE_{21}}{\hbar}\exp\left(\frac{iE_{21}}{\hbar}t\right)C_2 - \frac{i}{\hbar}H_{21}\exp\left(\frac{iE_{21}}{\hbar}t\right)\frac{\partial C_2}{\partial t}$$

$$\frac{\partial^2 C_1}{\partial t^2} = -\frac{i}{\hbar}H_{21}\exp\left(\frac{iE_{21}}{\hbar}t\right)\left[\frac{iE_{21}}{\hbar}C_2 + \frac{\partial C_2}{\partial t}\right]$$

$$\frac{\partial^2 C_1}{\partial t^2} = -\frac{i}{\hbar}H_{21}\exp\left(\frac{iE_{21}}{\hbar}t\right)\left[\frac{iE_{21}}{\hbar}\frac{\partial C_1}{\partial t}\left(-\frac{\hbar}{iH_{21}}\exp\left(-\frac{iE_{21}}{\hbar}t\right)\right) - \frac{i}{\hbar}H_{21}\exp\left(\frac{iE_{21}}{\hbar}t\right)C_1\right]$$

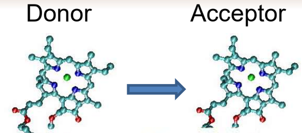
$$\frac{\partial^2 C_1}{\partial t^2} = +\left(\frac{i}{\hbar}H_{21}\frac{iE_{21}}{\hbar}\frac{\hbar}{iH_{21}}\right)\frac{\partial C_1}{\partial t} + \left(\frac{i}{\hbar}H_{21}\exp\left(\frac{iE_{21}}{\hbar}t\right)\frac{i}{\hbar}H_{21}\exp\left(\frac{iE_{21}}{\hbar}t\right)\right)C_1$$

$$\frac{\partial^2 C_1}{\partial t^2} - \frac{iE_{21}}{\hbar}\frac{\partial C_1}{\partial t} + \frac{H_{21}^2}{\hbar^2}\exp\left(\frac{2iE_{21}}{\hbar}t\right)C_1 = 0$$

← 2nd order diff eq, can solve for C_1

13

Time-dependent SE: solution



$$E_{21} = H_{22} - H_{11}$$

Solution for $C_1(0)=1$, $C_2(0)=0$:

$$\Omega = (1/2)\left[(E_{21})^2 + 4(H_{21})^2\right]^{1/2}$$

$$C_1(t) = \{\cos(\Omega t/\hbar) - i(E_{12}/2\Omega)\sin(\Omega t/\hbar)\}\exp(iE_{21}t/2\hbar)$$

$$C_2(t) = -i(H_{21}/\Omega)\sin(\Omega t/\hbar)\exp(-iE_{21}t/2\hbar)$$

Take a square for population dynamics:

$$|C_2(t)|^2 = (H_{21}/\Omega)^2\sin^2(\Omega t/\hbar) = (1/2)(H_{21}/\Omega)^2[1 - \cos(2\Omega t/\hbar)]$$

- Population on acceptor oscillates with period $T = 2\pi\hbar/2\Omega = h/2\Omega$
- Mean probability is $(H_{21}/\Omega)^2/2$

Special cases:

$E_{21}=0$ (homodimer) → $|C_2(t)|^2 = (1/2)[1 - \cos(2H_{21}t/\hbar)]$ $\quad T = h/(2H_{21})$

$|E_{21}|^2 \gg 4|H_{21}|^2$ → $|C_2(t)|^2 = 2(H_{21}/E_{21})^2[1 - \cos(E_{21}t/\hbar)]$ $\quad T = h/E_{21}$

Note: Ω is exactly the same as energy splitting in excitonic states (Lecture 15): $\pm\frac{1}{2}\sqrt{\delta^2 + 4(H_{12})^2}$

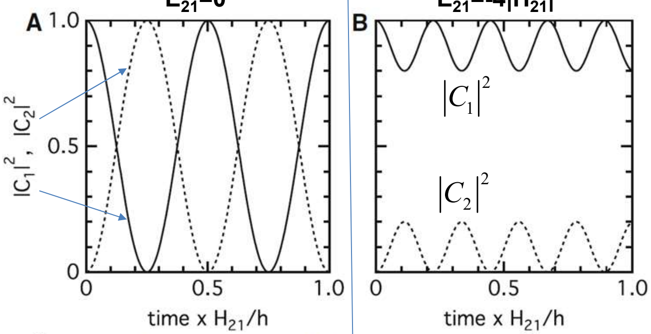
14

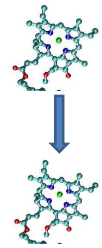
Population oscillations

Special cases: $(C_1(0)=1, C_2(0)=0)$

$$\Omega = (1/2)\left[(E_{21})^2 + 4(H_{21})^2\right]^{1/2}$$

$E_{21}=0$ **$E_{21}=-4|H_{21}|$**





$$|C_2(t)|^2 = (1/2)[1 - \cos(2H_{21}t/\hbar)]$$

Period: $2H_{21}T/\hbar = 2\pi$ $(h = 2\pi\hbar)$

$T = h/2\Omega = h/(2H_{21})$

Even with very small $H_{21,}$ the system will be entirely in state 2 at some times!
Larger H_{21} increases frequency

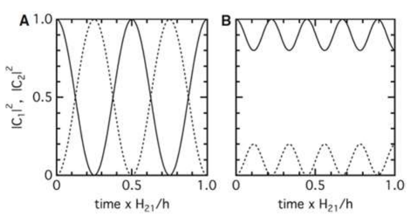
$$|C_2(t)|^2 = 2(H_{21}/E_{21})^2[1 - \cos(E_{21}t/\hbar)]$$

$T = h/2\Omega \approx h/E_{21}$

Larger E_{21} :
- Frequency increases
- Amplitude decreases
- The probability of transfer decreases

15

Philosophical question…

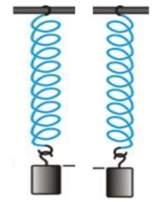


What is *__quantum effect__* and what is not?

Common perception: population beats between states is a quantum effect. Absorption of light, excitonic states, etc. are not really quantum, or not quantum at the same level as population beats…

16

Classical analog: spring-mass oscillators



Imagine two spring-mass oscillators.
For independent oscillators, motion is governed by Newton's law under Hook's spring force:

$$F = ma$$
$$-kx = ma \Rightarrow \ddot{x} = -\frac{k}{m}x$$

The solution for that equation is a harmonic motion:

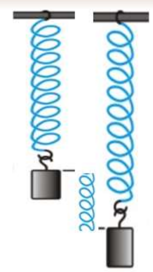
$$x = A\cos(\omega t) \qquad \omega = \sqrt{k/m} \qquad f = \frac{1}{T} = \frac{\omega}{2\pi}$$
$$v = \dot{x} = -A\omega\sin(\omega t) = -A\sqrt{k/m}\sin(\omega t)$$

And total energy at any time is conserved (kinetic + potential)

$$E = \frac{mv^2}{2} + \frac{kx^2}{2} = \frac{kA^2}{2}$$

17

Classical analog: spring-mass oscillators



Imagine two spring-mass oscillators.
Now, add a weak spring (interaction) between the two oscillators. In this case the force on mass m_1 and m_2 depends on the displacement from equilibrium for each mass (x_1 and x_2) as well as the separation between the two masses $x_{12}=(x_2-x_1)$ or $x_{21}=(x_1-x_2)$:

$$F_1 = -k_1x_1 - k_{12}x_{12} = m_1\ddot{x}_1$$
$$F_2 = -k_2x_2 + k_{12}x_{12} = m_2\ddot{x}_2$$

The system of equation becomes:

$$\ddot{x}_1 = -\omega_1^2 x_1 - \omega_{c1}^2 x_{12}$$
$$\ddot{x}_2 = -\omega_2^2 x_2 - \omega_{c2}^2 x_{21}$$

Where:

$$\omega_i = \sqrt{k_i/m_i}$$
$$\omega_{ci} = \sqrt{k_{12}/m_i}$$

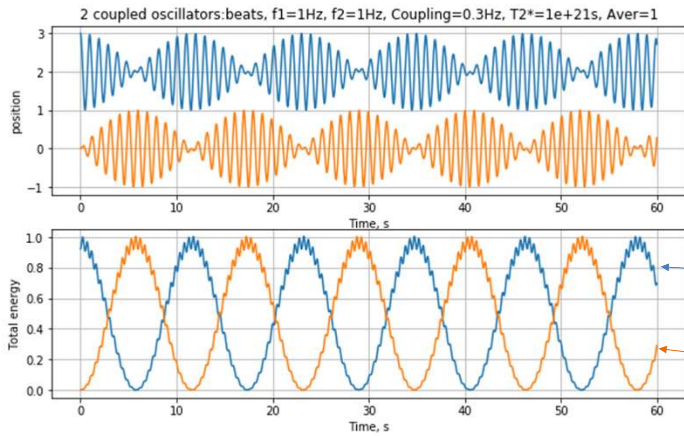
Note that Newton 3rd law is satisfied: $\omega_{c1}^2 x_{12} = -\omega_{c2}^2 x_{21}$

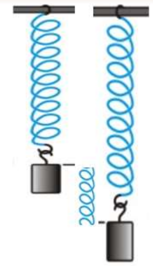
This system can be easily solved numerically

18

Classical analog: spring-mass oscillators

Simulation using a simple Python script:
T=0: spring 1 (blue) is stretched, spring 2 (orange) is at equilibrium
Resonance frequencies: $\omega_1=\omega_2$





$$\ddot{x}_1 = -\omega_1^2 x_1 - \omega_{c1}^2 x_{12}$$
$$\ddot{x}_2 = -\omega_2^2 x_2 - \omega_{c2}^2 x_{21}$$

$|C_1|^2$
$$E_1 = \frac{m_1 v_1^2}{2} + \frac{k_1 x_1^2}{2}$$

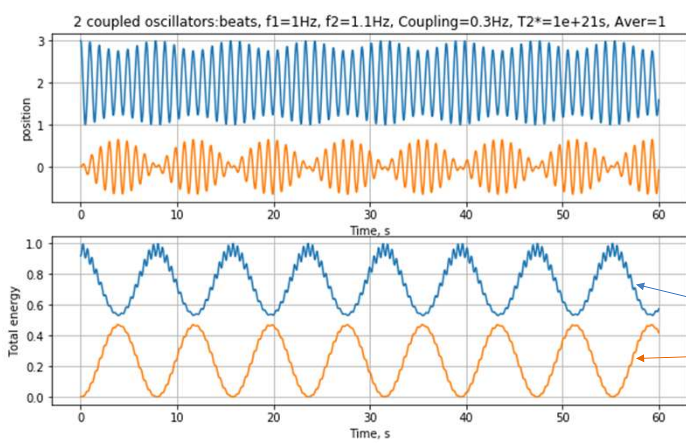
$|C_2|^2$
$$E_2 = \frac{m_2 v_2^2}{2} + \frac{k_2 x_2^2}{2}$$

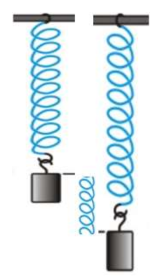
Q: Why there are fast oscillations in both energies?

19

Classical analog: spring-mass oscillators

Simulation using simple Python script:
T=0: spring 1 (blue) is stretched, spring 2 (orange) is at equilibrium
Resonance frequencies: $\omega_2=1.1\omega_1$





$$\ddot{x}_1 = -\omega_1^2 x_1 - \omega_{c1}^2 x_{12}$$
$$\ddot{x}_2 = -\omega_2^2 x_2 - \omega_{c2}^2 x_{21}$$

$|C_1|^2$

$|C_2|^2$

20

Classical coupled oscillators

Examples:

Coupled pendulums: https://youtu.be/M0Zjl_tMOZg?t=37

Wilberforce pendulum https://youtu.be/M0Zjl_tMOZg?t=105

See coupled pendulum simulations:
https://www.walter-fendt.de/html5/phen/coupledpendula_en.htm

21

Population oscillations: damping

Our theory is derived for a *single* system with sharply defined energies. The theory predicts that oscillations will persist forever!

In experiments, population oscillations exist and/or can be observed for a very short time (<1 ps):

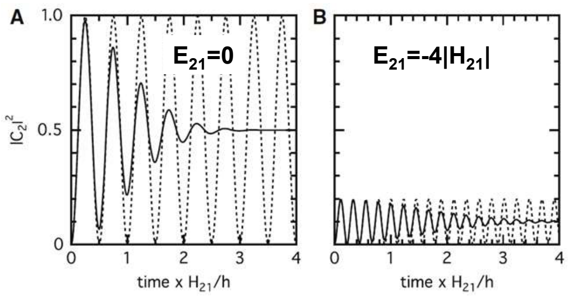
WHY?

22

Population oscillations: damping

1. Dephasing due to ensemble inhomogeneity

An experiment typically measures the ensemble average of many coupled oscillators
Different parts will oscillate at different frequencies, oscillations get out of phase and cancel → damped oscillations converging to the average population $<(H_{21}/\Omega)^2/2>$ *(<> means averaged over population)*



Dashed: ideal single system

Solid: averaged over uncorrelated Gaussian distributions of E_{21} and H_{21}. Distribution widths are $0.2H_{21}$ fwhm (full width at half maximum)

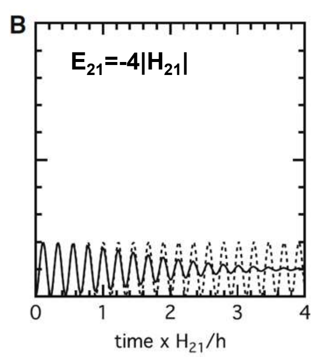
The oscillations of an ensemble are said to be *coherent* if the individual microscopic systems that comprise the ensemble oscillate in phase, and *incoherent* or *stochastic* if the phases of the microscopic systems are random

23

Population oscillations: damping

2. Coupling with the bath

If the system can transfer energy to or from the surroundings, it will relax toward an equilibrium in which the relative probabilities of being in various states are given by the Boltzmann equation.



← The system evolves to population C_2^2 ~ 0.1 even after long time.
However, since state 2 (excitation on acceptor) has lower energy ($E_{21}<0$) one expects the population of state 2 to evolve towards a Boltzmann distribution.
Our equations, however, have no temperature dependence; for $E_{21}>>kT$, the population of state 2 should approach 100%.

24

Population oscillations

3. Finite lifetimes of excited states

Homogeneous broadening applies to an ensemble of molecules and to individual donor-acceptor pairs as well → oscillations will be damped even for a single pair.

Förster's theory assumes energy transfer between states with different transition energies and explicitly accounts for the Boltzmann distribution. But it is silent about oscillations predicted by the above equations. It does not address how rapidly the system relaxes into an incoherent state, assuming it is fast compared with transfer rates. However, energy transfer between closely spaced pigments is often <<1 ps.

Need to understand how large ensembles of molecules interact (stochastically) with their surroundings.